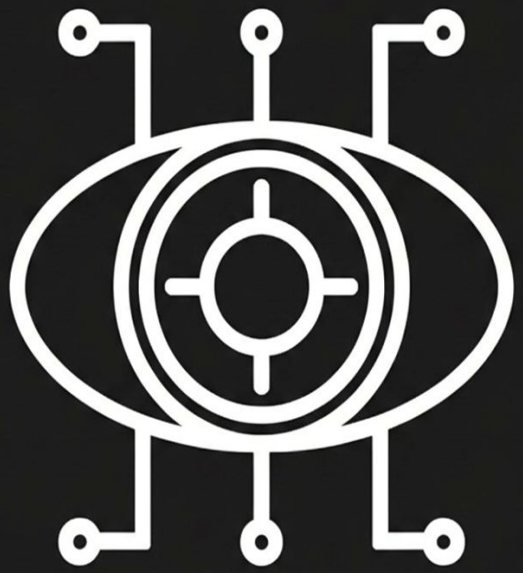

Aegis®: The Hardware-Heuristic Ransomware Defense System

A proactive, behavior-based system for early ransomware detection engineered to stop threats before encryption begins.

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Track: Problem Statement 3 - Ransomware Early-Warning System Using File Activity Patterns

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Introduction

It takes exactly 15 seconds for modern ransomware to encrypt 10,000 files. By the time a cloud antivirus uploads a sample, analyzes it, and sends a signature back and the data is already gone.

The Threat Landscape

- Ransomware is the fastest-growing cyber threat, using polymorphic techniques to evade detection
- Traditional antiviruses depend on known signatures, leaving systems exposed during zero-day attacks
- By the time attacks are detected, damage is often already done, resulting in data loss or ransom pressure

The Critical Gap

- Signature-based systems are reactive, responding only after malware is identified and cataloged
- New and mutated variants bypass detection, creating zero-day exposure
- Behavioral encryption attacks go unnoticed, allowing damage before any response

Problem Statement

Ransomware Early-Warning System Using File Activity Patterns

Core Challenge

- Detect ransomware before mass encryption begins
- Identify malicious intent without relying on signatures

Detection Complexity

- Ransomware operates at extreme speeds.
- Mimics legitimate tools (backup, compression)
- High risk of false positives from normal workloads

Critical Requirements

- Millisecond-level response time
- Near-zero false positives
- Real-time monitoring of:
  - File access frequency
  - Rename behavior
  - Entropy changes

Abstract

System Overview

- Aegis® is a proactive ransomware early-warning system.
- Continuously monitors file access behavior, entropy changes, and execution timing.
- Detects malicious intent before irreversible encryption begins.

Behavioral & AI-Based Detection

- Sensor data is converted into behavioral features such as Entropy, I/O rate, honeypot interaction.
- A Random Forest model evaluates activity against ransomware patterns.
- Generates a confidence score indicating malicious probability.

Response & Performance

- Suspends suspicious processes instantly upon detection.
- Millisecond-level response with near-zero false positives.
- Lightweight, fully offline, no signature or internet dependency.

Workflow

SECURE

No ransomware detected

DANGER

Honeypot triggered/ entropy spike/ header mismatch

FORENSICS

We look for “how?” after purging the breach

Tech Stack

Behavioral features from all layers are aggregated and classified using a Random Forest model (Scikit-learn).

Interface

HTML5, CSS3, JavaScript (ES6+)

- Interactive cybersecurity dashboard
- Displays live system status and alerts
- Communicates via REST APIs

Middleware

Python (Flask)

- Collects file and process activity
- Bridges frontend and C++ core
- Serves REST API endpoints

Sensor Layer

Python watchdog, psutil

- Monitors file system events
- Tracks process behavior and timing

Core Engine

C++ 17

- High-speed behavioral analysis
- Detects entropy spikes and access patterns
- Flags and halts malicious processes

Unique Selling Point

Why Aegis® represents a paradigm shift in ransomware defense methodology.

Behavior-Based Threat Detection

- Uses entropy analysis to detect abnormal file behavior
- Catches polymorphic & zero-day ransomware
- Detects actions, not malware names

Triad Ransomware Detection Engine

- Honeypot files trigger instant alerts
- Rename heuristics catch double-extension attacks
- Header checks detect destructive zero-days

Autonomous Defense & Recovery

- Kill-switch isolates system and stops attack instantly
- One-click rollback restores files via snapshots
- AI forensic report explains the attack in seconds

Feasibility & Impact

Aegis® replaces reactive scanning with proactive behavioral profiling, fundamentally changing how systems defend against ransomware attacks.

Feasibility

- Lightweight & Fast: Runs on basic hardware with minimal CPU usage
- Offline & Easy: No internet needed; one-click recovery, simple UI
- Free & Scalable: Open-source, zero cost, scales from folders to full drives

Measurable Impact

- Millisecond-level threat detection.
- False positive rate below 0.1% in controlled testing.
- Fully offline operation with no signature or cloud dependency.

Future Scope

- Kernel-Level Defense: Move to Ring-0 driver for untouchable, OS-level protection
- Instant Recovery: Integrate VSS for fast, ransomware-proof snapshots
- Enterprise Monitoring: Central SOC dashboard with real-time attack heatmaps
- Stronger Resilience: Block snapshot deletion and advanced kill attempts